Program : **Diploma in Printing Technology**	
Course Code : **5103B**	Course Title: **Maintenance Engineering & Quality Control**
Semester : **5**	Credits: **4**
Course Category: **Program Elective**	
Periods per week: **4 (L:4, T:0, P:0)**	Periods per semester: **60**

Course Objectives:

- Maintenance of printing machines is important for many reasons. The delay in production for an equipment failure can create serious problem because printing is a service industry.
- Like other technological fields, new concepts and applications are developing continuously in maintenance also. This proposed syllabus is based on latest changes.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Working of Sheet fed and Web fed offset technologies		Mechanism of Printing Machines-I	3
Working of Flexo and Gravure Printing		Mechanism of Printing Machines-II	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Define different maintenance programs, safety precautions and lubrication.	15	Understanding
CO2	Demonstrate different drives used for power transmission.	14	Understanding
CO3	Demonstrate machine erection and reconditioning in detail also understands electrical elements.	15	Understanding
CO4	Illustrate Quality control in printing industry.	14	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Define different maintenance programs, safety precautions and lubrication**.		
M1.01	Introduction to maintenance.	3	Remembering
M1.02	Different types of maintenance	4	Remembering
M1.03	Safety precautions and housekeeping.	4	Remembering
M1.04	Types of lubrications and lubricating methods in printing industry	4	Understanding
Contents: Maintenance - Definition, Objectives and functions. Types of Maintenance - Planned maintenance and unplanned maintenance. Planned maintenance - Types-Preventive Maintenance, Predictive Maintenance, and Scheduled maintenance. Unplanned maintenance - Breakdown Maintenance. Contract maintenance. - Preventive Maintenance - Advantages, Functions - Planning, scheduling - Repair cycles, Dispatching and Controlling. Categories - Inspection, Small repair, Medium repair and Overhauling. Safety Precautions and House Keeping - safety precautions to be followed in press area - Aids to good housekeeping. Lubrication - Introduction - Advantages - Types of Lubricants - Solid, Semisolid and Liquid. Lubrication Schedule - Chart and Paint Marks. Methods of Lubrication - Gravity feed method-Wick feed, oil feed and Drip Feed , Force feed or Pressure feed method - Oil can, grease gun and grease cup, Splash method - Ring oiler, and Continuous/Multipoint flow method.			
CO2	**Demonstrate different drives used for power transmission.**		
M2.01	Chain drives.	3	Understanding
M2.02	Belt drives.	2	Understanding
M2.03	Gear drives.	3	Understanding
M2.04	Bearings.	2	Understanding
M2.05	Cam and followers.	2	Understanding

M2.06	Springs.	2	Understanding
	Series Test I	1	
Contents: Chain Drives - Introduction, Types of Chains(Briefly) - Roller Chain, Silent Chain, Ewart Chain and Bead Chain - Merits and Demerits of Chain Drives. Belt Drives - Introduction, Types of Belts(Briefly) - Flat belt, Rope belt, Tooth Belt, V belt and Timing Belt- Merits and Demerits of Belt drives. Gear Drives - Introduction, Types of Gears(Briefly) - Spur gear, Helical gear, Bevel gear, Worm gears and Herringbone gear - Merits and Demerits of gear drives. Maintenance and Lubrication of Drive Systems - Chain Drive, Belt Drive, and Gear Drive. Direct drive technology - Introduction - Advantages - Application in the printing field. Bearings - Introduction, Merits and Demerits. Types of Bearings (Briefly) - Sliding bearings and Antifriction bearings - Ball bearings, Needle bearings and Roller bearing. Cams and Follower - Advantages of cam and Follower. Types of Cams and Followers (Briefly) - Disk Cam, Translation Cam, Groove Plate Cam, Cylindrical Cam, Eccentric Cam and Tow Wipe Cam. Springs - Types of springs (Briefly) - Helical Spring, Conical spring, Volute Spring and Torsion Springs and its application.			
CO3	**Demonstrate machine erection and reconditioning in detail also understand electrical elements.**		
M3.01	Machine erection.	5	Understanding
M3.02	Machine reconditioning.	5	Understanding
M3.03	Electrical drives.	5	Understanding
Contents: **Machine Erection and Reconditioning** - Introduction, Equipments and Tools used for it, and application. **Machine Erection** - Equipment needed for erection - selection of location and environmental conditions - erection procedure for various prepress. Printing and finishing equipments and machinery. **Reconditioning** - Principles of reconditioning-repair methods of various parts - roller copperising and rerubberising - ebonite covering- dampening and inking systems - paper transport systems - cylinder bearing supports - eccentrics. **Maintenance of electrical systems** - AC motors and DC motors- electromagnetic friction couplings. Electromagnets - magnetic starters and contractors - limit switches - knife switches - micro switches starting and regulating rheostat - electric panels - electrical apparatus and electric wiring on the machines.			
CO4	**Illustrate Quality control in printing industry.**		
M4.01	Quality.	1	Understanding
M4.02	Quality & Process control targets / patches.	4	Understanding
M4.03	Electrical system, pneumatic system and hydraulic system maintenance.	3	Understanding
M4.04	Quality/process standardization methods & techniques.	3	Understanding

M4.05	Trouble shooting & Importance of calibrating printing machine (offset).	3	Understanding
	Series Test II	1	

Contents:

Quality: ISO definition. Standardization - ISO definition. Need of quality control. Advantages of quality control/standardization. Quality assessment methods - subjective (visual), objective, measured. How a customer determine the quality. Customer satisfaction. Factors of influence and specifications determining the quality of the offset print. Introduction to Different process/quality/standard/material research firms/organization - ISO, GATF, FOGRA, UGRA, PIRA, Brunner, IGT, CIE, ICC, Printing Industries of America (PIA). Quality & process control targets / patches: GATF/Printing Industries Plate Control Target. UGRA/FOGRA media wedge. UGRA plate control target film. UGRA/FOGRA process control elements in. UGRA/FOGRA Digital plate control wedge. UGRA/FOGRA digital print scale. Color control Bars - parts of color control bar - tint patches, solid patches, grey patches, star targets, RGB overprints, total area coverage patches. GATF CTP printing plate exposure wedge. GATF Grey balance patch. GATF ICC color charts. GATF test forms - contains different targets and sizes. GATF star target. GATF line screen (visual and measured). Brunner zebra strip. Brunner color control strip. Brunner CTP plate control strip. Brunner CMYK marks. Registration patches for unit to unit registration. Quality/process standardization methods & techniques: Optimization. Lean printing definition, Advantages. Lean printing definition. Lean goals. 7 types of wastes in Lean. Lean tools - 5s, standardized work instructions, value stream mapping, Total proactive maintenance, Kaizen events, Just in time management, 5 whys, etc. Six Sigma - Six Sigma concept, definition, advantages. Troubleshooting and Importance of calibrating printing machine (offset).

Text / Reference:

T/R	Book Title/Author
T1	GATF Lithographers manual GATF , USA.
R1	Herschel L Apfelberg Maintaining printing equipment GATF.
R2	Barbera, L Albinini Solving web offset press problem GATF.
R3	HP Garg Industrial maintenance S Chand and company ltd.

Online resources:

Sl. No	Website Link
1	http://edelmann-printingmachines.com/
2	https://learnmech.com › mechanical-drives-belt-chain-gear
3	http://www.yourarticlelibrary.com/industries/maintenance-management